

Milestone 1 - Requirements Document

Team: Team 01 - SmashGo

Topic: D2 - Sports Platform

Members:

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Repository: <https://github.com/SE-FDA-NEU-AI66B/software-group1-topicd2.git>

Project board: <https://github.com/orgs/SE-FDA-NEU-AI66B/projects/24>

Submitted by: Chủ Vũ Thảo Hiền

All issues

is:issue state:open

Open 10

Closed 2

Author Labels Projects Milestones Assignees Types Newest

Open

Task

#20

dqchien opened 3d ago

Góp file, review chéo, xuất PDF nộp bài

submission

Open

Task

#19

dqchien opened 3d ago

README + docs/sprint-log.md

documentation

Open

Task

#18

dqchien opened 3d ago

Tạo 10 GitHub issue label "story" + gán Sprint 1

repo-setup

Open

Task

#17

dqchien opened 3d ago

10 User Stories + Acceptance Criteria

documentation

Open

Task

#16

dqchien opened 3d ago

Sprint 1 wrap-up

chore

Open

Task

#15

dqchien opened 3d ago

Screens & Flow Diagram

documentation

Open

Task

#14

dqchien opened 3d ago

Scenarios (2 kịch bản sử dụng)

documentation

Open

Task

#13

dqchien opened 3d ago

Refine backlog for Sprint 1

chore

Open

Task

#12

dqchien opened 3d ago

Business Rules (BR1-BR6)

documentation

Open

Task

#11

dqchien opened 3d ago

Product Vision & Personas (2 personas + phỏng vấn)

documentation

Sprint Board — Sports booking platform

Add status update

Insights

Workflows 7

View 1

New view

Filter by keyword or by field

View

Todo 10

This item hasn't been started

software-group1-topicd2 #29

[Story] US-01 — View Available Badminton Courts

software-group1-topicd2 #30

[Story] US-02 View Detailed Court Information

software-group1-topicd2 #31

[Story] US-03 - View Available Time Slots

software-group1-topicd2 #32

[Story] US-04 — Book One Available Time Slot

software-group1-topicd2 #33

[Story] US-05 — Prevent Double Booking

In Progress 0

This is actively being worked on

In Review 0

Done 13

This has been completed

software-group1-topicd2 #11

[Task] Product Vision & Personas (2 personas + phỏng vấn)

#22 Sprint 1

software-group1-topicd2 #19

[Task] README + docs/sprint-log.md

#19 Sprint 1

software-group1-topicd2 #12

[Task] Business Rules (BR1-BR6)

#23 #25 Sprint 1

software-group1-topicd2 #20

[Shared] Góp file, review chéo, xuất PDF nộp bài

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1. Product vision

SmashGo is a system that enables badminton players of all levels to book courts for their desired time slots and secure a reservation in just minutes. This eliminates the need for manual methods - such as calling or visiting the facility to speak with the manager - which fail to provide real-time availability updates and often lead to scheduling conflicts between groups.

2. Personas

Lê Sỹ Huy – 20-year-old third-year student and recreational badminton player.

- Plays badminton with friends about four times a week; usually the one who books the court for the group. Typically plays in the 8:00 PM – 10:00 PM slot at familiar venues like Maxping Vĩnh Tuy and Phúc Long Badminton Court.
- Goal:** To book courts faster, guarantee availability, and avoid missing a session due to a busy schedule.
- Blocked by:** Needs immediate access to comprehensive information—price, availability, court conditions, and location—to make quick decisions; also requires the ability to remind schedule the booking in advance (around 4 hours prior to play).
- In his words:** "I find the way the app I'm currently using (ALOBO) works to be satisfactory."
- Interview note:** Interviewed at 2:00 PM – 2:10 PM on September 16, 2026.

Chữ Vũ Thảo Hiền – 20-year-old third-year university student and recreational badminton player.

- Plays badminton with friends about three times a week; previously responsible for booking courts for the club. Usually plays from 3:00 PM to 5:00 PM at Đức Thảo Badminton Court (18 Tam Trinh). Before discovering the ALOBO app, I used to book courts by messaging the owner via Zalo, which often led to scheduling conflicts with other groups.
- Goal:** To book a court quickly, secure a confirmed reservation, and ensure there are no overlaps with other groups.
- Blocked by:** Needs access to comprehensive information—actual photos of the court, pricing, availability, court condition, and location.
- In my words:** "Booking a court used to be complicated and risky before I found the ALOBO app. However, since using the app, I've found the process to be smooth and trouble-free."
- Interview note:** Based on personal experience.

3. Scenarios

Scenario 1 — Lê Sỹ Huy: Quickly booking a familiar court

- Lê Sỹ Huy decides to play badminton with his friends at 8:00 PM and needs to arrange the court before the evening.
- He searches for courts that are available at the desired time and checks the basic information, including price, location, and court condition.
- Since he usually plays at familiar venues, he chooses a court he has played at before.
- He selects the 8:00 PM–10:00 PM time slot and verifies that it is available.
- He provides the required information and confirms the reservation.
- The system records the booking and provides the confirmed reservation details. 7. Huy schedules a reminder for around four hours before the session so he can remember the booking despite his busy schedule.

7. Before leaving for the session, he checks the reminder and reservation details, then goes to the court with his friends.

Scenario 2 — Chủ Vũ Thảo Hiền: Finding a suitable court and avoiding booking conflicts

1. Chủ Vũ Thảo Hiền needs to arrange a badminton court for her group for an upcoming session from 3:00 PM to 5:00 PM.
2. Because she previously experienced scheduling conflicts when booking through Zalo, she first looks for courts with clearly confirmed availability.
3. She examines several available options and checks their actual photos, prices, court conditions, and locations before making a decision.
4. She finds a suitable court at Đức Thảo Badminton Court and checks whether the desired 3:00 PM–5:00 PM period is available.
5. After confirming the available time, she enters the required information and submits the reservation.
6. The system confirms the reservation and provides the booking details, including the selected court and time.
7. Hiền checks the confirmation to make sure her group's reservation has been successfully recorded and does not overlap with another booking.
8. On the scheduled day, she and her friends arrive at the reserved court and play at the confirmed time.

4. User Stories

ID	Story	Priority	Points
US-01	As a user, I want to view available badminton courts so that I can find a suitable court for my session.	P0	3
US-02	As a user, I want to view detailed court information, including price, photos, location, and court condition, so that I can choose a suitable court.	P0	3
US-03	As a user, I want to view available time slots for a court so that I can select a suitable playing time.	P0	3
US-04	As a user, I want to book one available time slot so that I can secure a badminton court for my group.	P0	5
US-05	As a user, I want the system to prevent double booking so that my reservation does not overlap with another user's booking.	P0	5
US-06	As a user, I want to cancel my booking so that I can release a court when I no longer need it.	P1	3
US-07	As a user, I want to receive a booking reminder so that I do not forget my scheduled badminton session.	P1	3
US-08	As a user, I want to view my booking history so that I can check my previous and upcoming reservations.	P1	3
US-09	As a user, I want to check in for my booking so that the system can record that I have arrived and used the court.	P1	3

ID	Story	Priority	Points
US-10	As an administrator, I want to manage badminton courts and view usage frequency by time slot so that I can keep court information up to date and understand booking patterns.	P2	5

US-01 — View Available Badminton Courts

Acceptance Criteria

1. **Given** badminton courts are available at 8:00 PM,
When the user searches for courts at 8:00 PM,
Then the system displays the available courts for that time.
2. **Given** a court has no available slots at 8:00 PM–10:00 PM,
When the user searches for courts during this period,
Then the system does not display that court as available.

US-02 — View Detailed Court Information

Acceptance Criteria

1. **Given** a badminton court has information including price, photos, location, and court condition,
When the user opens the court details,
Then the system displays all four types of information.
2. **Given** the selected court costs 100,000 VND per hour,
When the user views the court details,
Then the system displays the price as 100,000 VND per hour.

US-03 — View Available Time Slots

Acceptance Criteria

1. **Given** a court is available from 3:00 PM to 5:00 PM,
When the user views the court schedule,
Then the system displays the 3:00 PM–5:00 PM time slot as available.
2. **Given** the 8:00 PM–10:00 PM time slot has already been booked,
When the user views the court schedule,
Then the system displays the 8:00 PM–10:00 PM slot as unavailable.

US-04 — Book One Available Time Slot

Acceptance Criteria

1. **Given** the 8:00 PM–10:00 PM time slot is available,
When the user confirms the booking for that slot,
Then the system creates exactly one booking with the status "Confirmed".

2. **Given** the user has selected one available 2-hour time slot,
When the user submits the required booking information,
Then the system reserves only the selected court and time slot for that user.

US-05 — Prevent Double Booking

Acceptance Criteria

1. **Given** Court A is already booked from 3:00 PM to 5:00 PM on the selected date,
When another user attempts to book Court A for the same 3:00 PM–5:00 PM period,
Then the system rejects the second booking and keeps only one confirmed booking for that slot.
2. **Given** two users submit booking requests for the same court and the same 2-hour time slot at the same time,
When the system processes both requests,
Then exactly one request is confirmed and the other request is rejected.

US-06 — Cancel a Booking

Acceptance Criteria

1. **Given** the user has a confirmed booking from 3:00 PM to 5:00 PM,
When the user cancels the booking,
Then the booking status changes to "Cancelled".
2. **Given** a booking for the 3:00 PM–5:00 PM slot has been cancelled,
When another user views the court schedule,
Then the 3:00 PM–5:00 PM slot is displayed as available for a new booking.

US-07 — Receive a Booking Reminder

Acceptance Criteria

1. **Given** the user's badminton session starts at 8:00 PM,
When the user sets a reminder for 4 hours before the session,
Then the system schedules the reminder for 4:00 PM on the same day.
2. **Given** the user has cancelled a booking before the scheduled reminder time,
When the reminder time is reached,
Then the system does not send a reminder for the cancelled booking.

US-08 — View Booking History

Acceptance Criteria

1. **Given** the user has made 5 bookings,
When the user opens the booking history,
Then the system displays all 5 bookings associated with the user's account.
2. **Given** the user's bookings have statuses such as "Confirmed", "Cancelled", and "Completed",
When the user views the booking history,

Then the system displays the correct status for each booking.

US-09 — Check In for a Booking

Acceptance Criteria

1. **Given** the user's booking starts at 8:00 PM,
When the user checks in at 7:50 PM,
Then the system records the booking as "Checked-in".
2. **Given** the user does not have a confirmed booking for the selected court and time,
When the user attempts to check in,
Then the system rejects the check-in and does not create a check-in record.

US-10 — Manage Courts and View Usage Frequency

Acceptance Criteria

1. **Given** an administrator creates a badminton court named "Court A01",
When the administrator saves the court information,
Then the system creates "Court A01" as an active bookable court.
2. **Given** the system has at least 7 days of booking records,
When the administrator views usage statistics,
Then the system displays the number of bookings grouped by time slots, such as 8:00 PM–10:00 PM.

5. Business Rules

ID	Rule	Worked example
BR1	A court cannot be double-booked for a time slot that is already reserved.	Court 3 (Maxping Vinh Tuy) is already booked for 20:00–22:00. Another user tries to book the same court for 20:00–21:30 → rejected; the system shows "Court 3 is already booked for the 20:00–22:00 slot."
BR2	Every booking must last at least 1 hour. If multiple groups play on the same court back-to-back, their bookings must be contiguous — no gap is allowed between them.	Booking 20:00–21:00 → valid (meets the 1-hour minimum). On Court 1: Group A books 16:00–18:00, Group B books 18:30–... → invalid, because there is a 30-minute gap between the two bookings.
BR3	The slot is held for 10 minutes pending payment; the reservation will be automatically cancelled if this time limit is exceeded.	If a user selects Court 3 for the 20:00–21:00 slot at 18:00, the system holds the reservation until 18:10. If payment is not transferred by 18:10, the slot is automatically released and becomes available for others to book.
BR4	Cancelling more than 4 hours before play time gets a full refund; cancelling within 4 hours forfeits 50% of the deposit.	Booking for 20:00, cancelled at 15:00 (5 hours before) → full refund. Cancelled at 17:00 (3 hours before) → only 50% refunded.

ID	Rule	Worked example
BR5	The system automatically sends a reminder notification 4 hours before play time.	Booking starts at 20:00 → reminder notification is sent at 16:00 the same day.
BR6	Each account may hold at most 2 active (upcoming) bookings at a time.	A user already has 2 upcoming bookings (e.g., Sep 16 and Sep 18). Trying to add a 3rd booking → rejected with the message "You already have 2 active bookings — please complete or cancel one before booking another."

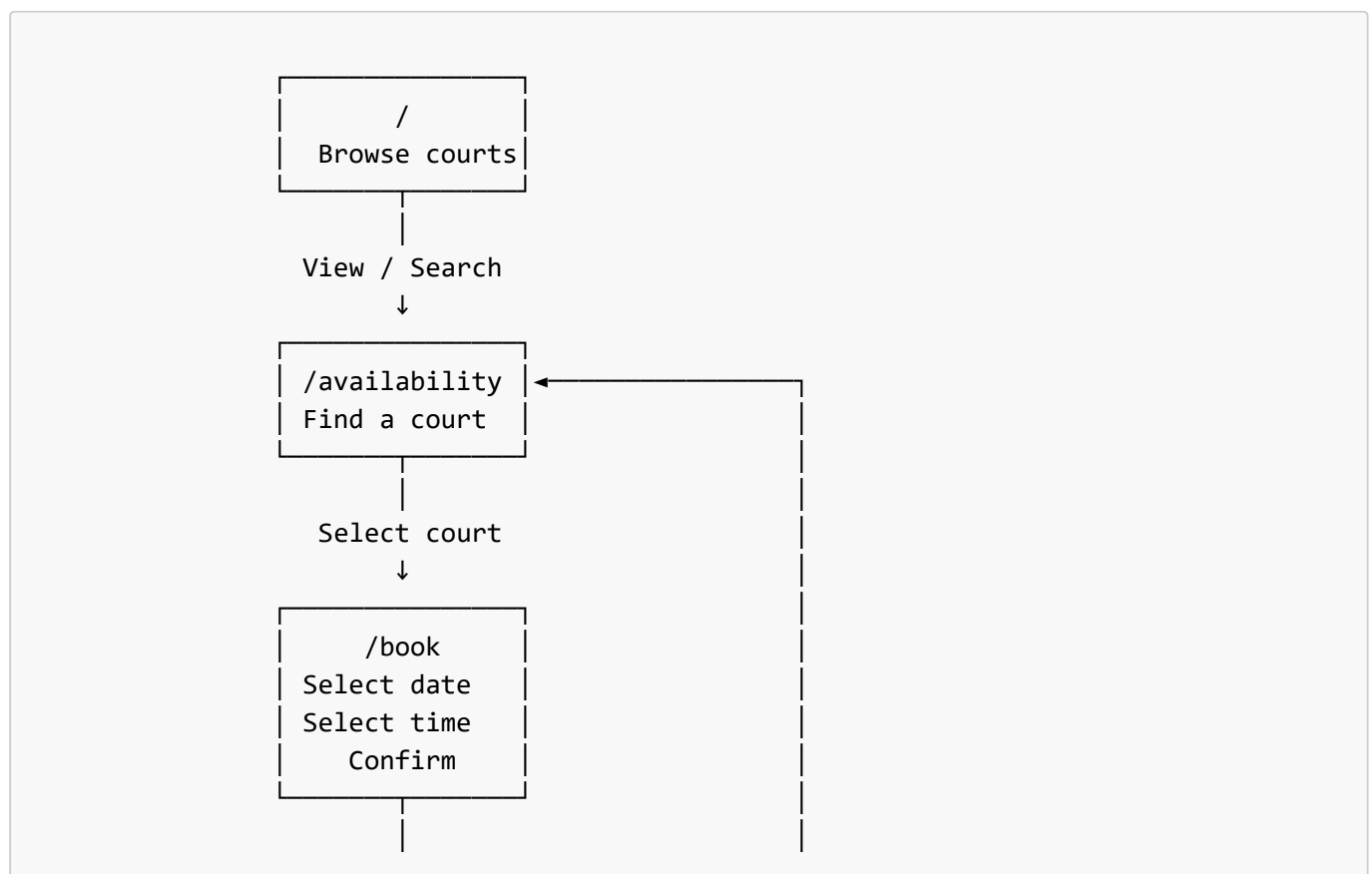
6. Screens and Navigation Flow

6.1 Screen Table

Route	Purpose	Access	Priority
/	Browse badminton courts and view court information	G, U	P0
/availability	View available courts by date and time	G, U	P0
/book	Select a court, date, and time, then proceed to booking/payment	G, U	P0
/my-bookings	View and cancel personal bookings	U	P0
/admin/courts	Add, edit, or disable badminton courts	A	P2

Access: G = Guest, U = User, A = Admin

6.2 Flow Diagram



Proceed to payment



Sign in
if Guest



Payment



/my-bookings
View / Cancel



Book again



/admin/courts
Add / Edit /
Disable courts



Update courts



/availability